AHMET FURKAN KURT

Engineering Project Portfolio — Architectural Decisions, Systems Design & Production Implementations

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> **Portfolio Purpose & Engineering Philosophy:** This document supplements my résumé by detailing the concrete architectural decisions, design patterns, and engineering trade-offs behind my core projects. My core philosophy is that user interfaces, game engines, and backends should not be treated as isolated silos, but designed together as unified, resilient systems—combining domain-driven data models, deterministic event loops, low-latency telemetry, and containerized cloud deployment. Projects below are organized by architectural depth and technical complexity.

1. ENGINE ARCHITECTURE, HUMAN-MACHINE INTERFACES (HMI) & SIMULATION

Temple Engine — Browser-Based 3D Game Engine & Visual Editor

CAPSTONE THESIS • 2026

`TypeScript` `Three.js / WebGL` `React` `Zustand` `cannon-es` `ECS Architecture` `EventBus` `Vite`

A full-featured 3D game engine core and visual editor running entirely in the browser, built from scratch with modern web standards and inspired by Unity's productive workflow. Rather than a superficial demo, it is an exercise in decoupled engine architecture where rendering is strictly isolated behind an adapter layer.

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Architecture: [React + Zustand Editor UI] ──API/Events──▶ [Pure TS Engine Core (ECS + EventBus + Loop)] ──Adapter──▶ [Three.js / WebGL]
```

- **Renderer-Agnostic Engine Core:** Implemented Scene, Entity, Component, and System hierarchies in pure TypeScript with zero runtime dependency on Three.js. Three.js is positioned strictly as a swappable renderer adapter responsible only for Object3D synchronization, camera mapping, and WebGL draws.
- **Deterministic Fixed-Timestep Loop:** Decoupled variable frame rendering from simulation physics using an accumulator-based delta/fixed-timestep loop, preventing tunneling and jitter in cannon-es rigid-body collision simulations.
- **Visual IDE & State Management:** Built an interactive editor with React and Zustand featuring a live scene hierarchy tree, 3D viewport with transformation gizmos (Translate/Rotate/Scale), dynamic component inspector, GLB asset loading, and Play/Stop modes with automated scene snapshot memory restoration.
- **Engineering Outcome:** Delivers desktop-grade developer ergonomics in a web environment; serializes scenes into clean, human-readable JSON capable of packaging into desktop binaries (via Electron/Tauri) or Android APKs.

Real-Time Hand Gesture Recognition & HMI Input System

COMPUTER VISION & HMI • 2026

— Computer Vision-Based OS Controller

`Python` `OpenCV` `MediaPipe (3D Landmarks)` `Linux X11 (xdotool)` `Tkinter Overlay HUD` `HMI Architecture`

A low-latency, modular Human-Machine Interface (HMI) pipeline that tracks 3D hand joints from a standard webcam feed in real time and translates biometric gestures into native operating system keyboard and mouse events. Fully documented with an accompanying academic research paper.

- **3D Joint Tracking & Palm Orientation Algorithm:** Analyzes MediaPipe 3D landmark coordinates to calculate palm normal vectors in real time, automatically toggling between Typing Mode (palm facing forward) and Mouse Navigation Mode (palm facing backward).
- **Chorded T9 Finger Typing State Machine:** Designed a temporal state machine detecting thumb-to-finger tap counts (1-4 taps) with calibrated timing windows (`COMMIT_IDLE_MS`, `TAP_WINDOW_MS`), enabling full alphanumeric text entry and two-hand number counting.
- **OS-Level Execution & Telemetry HUD:** Emits native Linux X11 inputs via `xdotool` directly to focused desktop applications with sub-frame latency. A transparent Tkinter on-screen display (Overlay HUD) renders live pinch thresholds and telemetry without stealing window focus.
- **Industry Relevance:** Offers hands-free, touchless control suitable for sterile medical consoles, aerospace ground stations, and industrial operator environments.

VR Earthquake Drill & 3D Interactive Simulation

VR & PERFORMANCE OPTIMIZATION • 2025–2026

— emeltek yazılım

Unity | C# | Virtual Reality (VR) | Custom Shaders / GLSL | Render Pipeline Optimization | Physics Simulation

An interactive 3D Virtual Reality training application that simulates realistic seismic tremors, structural debris hazards, and procedural evacuation protocols while measuring trainee response times.

- **Object-Oriented Physics & Scenario Logic:** Designed procedural tremor controllers, center-of-mass shifts for falling objects, and operator safe-zone validation algorithms in C# following OOP principles.
- **Critical Performance Boost (24 FPS → 60 FPS):** To eliminate VR simulation sickness, profiled and resolved GPU/CPU bottlenecks by combining meshes, minimizing draw calls, configuring static occlusion culling, and authoring custom lighting shaders—achieving a consistent 60 FPS (+150% improvement).

2. ENTERPRISE BACKEND ARCHITECTURE, DISTRIBUTED SERVICES & DATA INTEGRATION

Boutique Lavinia

ENTERPRISE FULL-STACK ARCHITECTURE

— Modular E-Commerce & Automated Legacy Migration Engine

.NET 8 | ASP.NET Core | PostgreSQL | Entity Framework Core | IdentityServer4 / JWT | Next.js | Docker Compose | Nginx

A production-grade, modular monolith e-commerce platform addressing the digital modernization of an active retail business. The system's crowning engineering achievement is an automated ETL pipeline that performed a seamless migration from a legacy vendor with zero data loss.

- **Legacy Ticimax ETL Migration Pipeline:** Architected a fault-tolerant import engine that parsed corrupt, inconsistently formatted Excel exports from the incumbent provider. Cleaned, normalized, and mapped 2,000+ SKU product catalogs, multi-tier hierarchical category trees, brand relationships, and customer records directly into relational PostgreSQL with 100% integrity.
- **Modular Monolith & Security:** Fully isolated catalog, inventory, order processing, and invoicing modules behind well-defined domain boundaries. Enforced authentication and token issuance via IdentityServer4 and JWT.
- **3-Tier Container Deployment:** Packaged Nginx (reverse proxy with SSL termination), ASP.NET Core API, and a Next.js storefront within Docker Compose, enabling deterministic, single-command zero-downtime rollouts.

NewsEngine (fikrimce.net)

BACKEND & DOMAIN-DRIVEN DESIGN

— Clean Architecture & CQRS Enterprise News Backend

.NET 10 | Clean Architecture | Domain-Driven Design (DDD) | CQRS | MediatR | FluentValidation | Serilog | Swagger / OpenAPI | Astro

A highly structured backend engine built for a journalism education publishing portal, implementing textbook Clean Architecture, Domain-Driven Design (DDD), and explicit editorial life-cycle governance.

- **CQRS via MediatR Pipeline:** Separated write operations (Commands: draft submission, editorial review, approval) from read operations (Queries: cached news feeds, metrics) into isolated handlers, ensuring high testability and single-responsibility compliance.
- **Explicit Editorial Domain Life-Cycle:** Implemented domain-level state machines governing article lifecycles (Draft → In Review → Approved → Published) alongside role-based access control (RBAC) and documented GDPR/KVKK compliance.
- **Validation, Structured Logging & Docs:** Filtered malformed inputs before reaching domain services using FluentValidation pipeline behaviors. Integrated Serilog structured logging and exposed 100+ fully documented RESTful endpoints via Swagger/OpenAPI.

Ashe's Touch (eyeOfTheBeholder)

SYSTEMS & GAME PROGRAMMING (RUST) • 2026

— 1.44 MB Size-Constrained Retro Adventure Game

Rust 2021 | Macroquad | Custom quad-snd Patch | LTO (Link-Time Optimization) | UPX Compression | Linux X11/PipeWire/ALSA | Windows Cross-Compile

A systems programming project built for the 1.44 MB Game Jam, where the entire standalone executable—including graphics, sound generation, level geometry, and game engine—strictly remains under 1,474,560 bytes (the size of a 3.5" floppy disk).

- **Aggressive Binary Footprint Optimization:** Configured release profiles with `opt-level = "z"`, fat Link-Time Optimization (`lto = "fat"`), single codegen units, panic abortion, and symbol stripping, compressed via UPX to compress the entire engine and assets under the megabyte barrier.
- **Procedural In-Memory Audio Synthesis:** Rather than embedding heavy audio waveforms, patched the `quad-snd` crate to generate algorithmic 8-bit sound effects procedurally at runtime using mathematical sine/square synthesis with 0 KB file footprint.
- **Dual-Target Cross-Compilation:** Developed natively on Void Linux (X11, ALSA, PipeWire) with automated MinGW cross-compilation scripts (`CROSS_WINDOWS=1`) delivering verified dual-platform release binaries.

3. AI TOOLING, CROSS-PLATFORM MOBILE & PRODUCTION WEB PLATFORMS

LLM Code Evaluation Dashboard & Benchmarking Workstation — llm-eval

R&D & MLOPS TOOLING

Python | FastAPI | Streamlit | SQLite | Kotlin & Flutter Toolchains | Detekt Static Analysis | Benchmarking Pipeline

An automated research and evaluation workstation that benchmarks generative LLM outputs against identical coding prompts by running them through real compiler toolchains and static analysis suites.

- **Objective Compiler-Driven Scoring:** Rather than relying on subjective text embeddings, executes headless Kotlin and Flutter compiler tools and Detekt static analysis rules to measure compilability, code smells, and instruction adherence.
- **FastAPI & Streamlit Suite:** Created an interactive workbench to trigger batch evaluations across models, archiving metrics in SQLite and graphing historical model drift.

Blog Thumbnail Generator — AI Media Pipeline — thumbnail-generator

AI INTEGRATION & MONOREPO

Astro (Node Adapter) | .NET 10 API | OpenRouter | Replicate (Flux) | Tailwind CSS | Monorepo Architecture

A full-stack monorepo application that automates the generation of 16:9 featured blog artwork directly from raw article text using chained generative AI models.

- **Secure Server-Side AI Proxy:** Insulates browser clients from third-party AI keys; the .NET 10 backend validates aspect ratios (`SendAspectRatio: 16:9`), enforces strict CORS, and handles upstream rate limits.
- **Two-Stage Prompt-to-Image Pipeline:** Converts Turkish article prose into structured English visual descriptions via OpenRouter (Gemini) before rendering high-fidelity imagery via a fine-tuned Flux diffusion model on Replicate.

renazteknik.com — High-Performance Commercial Platform

100/100 LIGHTHOUSE & SEO

— Live Business Product

Astro | Tailwind CSS | Firebase Firestore | Google Ads Conversion | Technical SEO

An active corporate customer acquisition site for an engineering and plumbing firm in Samsun, engineered to convert organic and paid search traffic into booked service appointments.

- **Flawless 100/100 PageSpeed & Technical SEO:** Delivered instantaneous page loads with zero extraneous JavaScript runtime using Astro, achieving perfect 100 scores across Performance, Accessibility, Best Practices, and SEO (+500% organic traffic surge).
- **Conversion Engine & Dynamic Reviews:** Integrated emergency call-to-action handlers, Google Ads conversion tracking, and live customer reviews backed by Firebase Firestore.

Flutter Cross-Platform Mobile Applications — Herkesyesin & libreQuiz

MOBILE ENGINEERING

Flutter | Dart | Riverpod / Provider | Firebase (Firestore, Auth, Storage) | RESTful API

- **Herkesyesin (Multi-Vendor Food Delivery Platform):** Developed at 208 Software Solutions; refactored 50+ feature modules with Clean Code and Riverpod state management to reduce runtime crashes by 70%, integrating live courier tracking and touch UX.
- **libreQuiz (Multiplayer Real-Time Quiz Platform):** Built during Sia Teknoloji internship; engineered a room-based multiplayer state engine synchronizing questions and scoring across mobile and web via Firebase.

4. COMPLEMENTARY SYSTEMS, TOOLING & OPEN SOURCE

ceylinOverlay — Linux X11 Desktop Video Overlay

Python, PyQt5, OpenCV (headless), python-xlib: A lightweight, hardware-accelerated video loop overlay utility rendering over the active X11 desktop manager during livestreams and presentations.

freelancer-finder — Rule-Based Opportunity Scoring CLI

Python, REST API, Pytest: An automated CLI and unit-tested scoring pipeline that parses API job feeds, ranking opportunities based on skill alignment, budget thresholds, and competition density.

MiniLan — Compiler Design & Lexical Analyzer

Lexical Analysis, Java, Automata Theory: An academic compiler engineering project implementing deterministic finite automata to tokenize grammar, parse syntax, and generate abstract syntax trees (AST).

Linux Environment & Browser Automation Scripts

Void/Arch Linux, i3wm, qutebrowser: A documented minimalist desktop environment (rice) alongside published open-source automation scripts including `qutebrowserRES` and `Watch Later`.

5. COMPREHENSIVE TECHNICAL MATRIX

Languages & Core Principles
C#, TypeScript, JavaScript (ES6+), Dart, Python, Rust, C++, Java, SQL, Bash, OOP

User Interfaces (UI/HMI) & Web
React, Next.js, Astro, Flutter, Tailwind CSS, PyQt5, Tkinter, Streamlit, HTML5/CSS3

Backend, Architecture & Distributed Systems
.NET 8/10, ASP.NET Core, EF Core, Clean Architecture, DDD, CQRS, MediatR, FastAPI, RESTful

3D, Simulation, Physics & Game Engines
Unity (C#, VR, Shaders), Three.js, WebGL, cannon-es, ECS, Macroquad (Rust), Phaser.js

Computer Vision, AI & Databases
OpenCV, MediaPipe 3D, PostgreSQL, SQLite, Firebase Firestore, LLM Benchmark, Flux

Systems, DevOps, Testing & Methods
Linux (Void/Arch, X11, PipeWire), Docker, Nginx, Git, CI/CD, Agile / Scrum, TDD, Detekt